

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 3: VHF Omnidirectional Range (VOR) & Doppler DVOR

Syllabus Source: CAE Oxford (pp. 89-130) | Pass Target: >= 90% AviationExam

1. VOR OPERATING PRINCIPLE & FREQUENCIES

VOR operates in the VHF band from 108.000 to 117.950 MHz (50 kHz channel spacing). It transmits 360 magnetic radials originating from the station:

- **Phase Comparison Principle:** The VOR transmits two separate 30 Hz signals: (1) Reference Signal: 30 Hz FM omnidirectional pulse transmitted simultaneously in all directions; (2) Variable Signal: 30 Hz AM directional beam rotating clockwise at 30 revolutions per second (1,800 rpm).
- **Radial Determination:** The phase difference between the Reference and Variable signals EXACTLY EQUALS THE MAGNETIC BEARING (RADIAL) from the station! At Magnetic North (000°), the signals are in phase (0° phase difference); at 090° Radial, phase difference is 90°.

2. CONVENTIONAL VOR (CVOR) VS DOPPLER VOR (DVOR)

Conventional VOR suffers from site error caused by ground reflections (buildings, mountains, trees) producing 'scalping' (needle oscillation) and course bends:

- **Doppler VOR (DVOR) Superiority:** DVOR uses a wide-aperture circular antenna array (~13.5 m diameter, 50 antennas) simulating counter-clockwise rotation at 30 revs/sec. The simulated motion creates a 30 Hz FM Doppler shift. Because of the wide aperture, DVOR is VIRTUALLY IMMUNE TO SITE REFLECTION ERRORS!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Reading VOR Radial Deflection &

Golden Rules of VOR Navigation: - Each dot on a standard CDI scale represents 2.0° of angular deviation (full 5-dot scale = 10°). - TO/FROM Flag: Indicates whether flying the selected OBS course will take you TO or FROM the VOR. Practical Example: - Aircraft is on Radial 090° outbound from VOR. - If OBS is set to 090°: Needle is centered, Flag reads FROM. - If OBS is set to 270° (inbound track): Needle is centered, Flag reads TO. - Reverse Sensing Trap: If flying outbound 090° with OBS set to 270° TO, the needle senses in reverse!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Reading Course Deviation Indicator (CDI)

SCENARIO (Standard EASA VOR Instrument Drill): An aircraft is flying on Magnetic Heading 360° (due North): - The pilot tunes the VOR and selects Omni-Bearing Selector (OBS) = 270° - The Course Deviation Indicator (CDI) needle deflects 3 DOTS TO THE RIGHT - The indicator displays a 'FROM' flag - Standard VOR CDI sensitivity: Full-scale deflection (5 dots) = 10° (2° per dot) QUESTION: On which radial is the aircraft currently located? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Determine the Angular Displacement from the needle deflection: - Needle is deflected 3 dots to the RIGHT. - Rule: 1 dot = 2.0 degrees. - Displacement = 3 dots x 2.0°/dot = 6.0° off course. Step 2: Understand the CDI with a 'FROM' Flag: - With a FROM flag and OBS selected to 270°: - Center needle = You are on Radial 270°. - Needle deflected RIGHT = The selected radial (270°) is to your RIGHT. - Therefore, your current position is to the LEFT of Radial 270°! - (Remember: On a compass rose, moving left/counter-clockwise from 270° means LOWER numbers: 270° - 6° = 264°). Step 3: State the Current Aircraft Radial: - Current Radial = 270° - 6° = Radial 264°! FINAL ANSWER: The aircraft is located on Radial 264° from the VOR station.

[!] EXAM TRAP (AVIATIONEXAM) - Reverse Sensing VOR Trap

- If you fly TOWARDS a VOR with a FROM flag selected: - The CDI gives REVERSE SENSING (if the needle is left, the track is to your right)! - To prevent reverse sensing: When flying inbound, ALWAYS select the inbound course with a TO flag!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the phase relationship between the reference signal and variable signal of a VOR on Radial 090°?**

- [A]: In phase (0°)
- [B]: The variable signal lags the reference signal by EXACTLY 90 DEGREES
- [C]: 180° out of phase
- [D]: 270°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The phase difference between the reference and variable 30 Hz signals matches the radial angle from magnetic north: on radial 090°, phase difference is 90°.

Q2: Why is a Doppler VOR (DVOR) significantly more accurate than a Conventional VOR (CVOR) in mountainous terrain?

- [A]: It uses UHF frequencies
- [B]: Its wide-aperture simulated rotating antenna array greatly REDUCES SITE AND REFLECTION ERRORS (scalloping)
- [C]: It uses satellite links
- [D]: It has higher transmitter power

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. DVOR's 13.5 m wide aperture antenna array creates a signal that is remarkably resilient against terrain and building multipath reflections.

Q3: On a standard VOR Course Deviation Indicator (CDI), what angular displacement is represented by FULL SCALE deflection?

- [A]: 5°
- [B]: 10 DEGREES (2.0° per dot)

- [C]: 2.5°
- [D]: 20°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Standard VOR CDI sensitivity is 2.0° per dot, meaning full-scale 5-dot deflection represents a 10° angular displacement off track.